

- 39 Which statement explains why alpha-particles have discrete energies, but beta-particles have a continuous range of energies?
- A Beta-particles have a much smaller mass than alpha-particles which means beta-particles can be emitted with a larger range of velocities.
 - B Only alpha-particles experience repulsion from the nucleus which is dependent on the number of protons in the nucleus.
 - C Only beta-particles are emitted with another lepton and some energy is transferred to the other lepton.
 - D Only the energy of alpha-particles is discrete because the composition of alpha-particles is always the same.
- 40 A hadron consists of antiquarks that are all identical.

What is a possible value, in terms of the elementary charge e , for the charge on the hadron?

- A $0e$ B $+1e$ C $+2e$ D $+3e$

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